PREVENTATIVE MAINTENANCE CHECKLIST

Enclosed Cab	Underground Haulage Truck	Tier 4	Dump

500	Hour

DATE: _____

UNIT NUMBER: _____

TECHNICIANS: _____

HOURS: _____

WORK ORDER: _____

***** PM TO BE COMPLETED ONCE THE UNIT IS WASHED *****

Note: All specifications, procedures, and parts not identified in this PM can be found in the OEM manuals

PRE-WORK SAFETY TASKS

MANDATORY PPE

LOCKOUT / TAGOUT – Ensure all procedures are followed to eliminate stored energy

Before working on brake system or hydraulic system drain accumulator pressure

Ensure deck, walkways and steps are free from foreign material such as oil and debris and that traction surface is in place and in good condition

Read and understand all warning plates on machine before operating, lubricating or repairing machine

To avoid burns, be alert for hot parts on machines, which have just been stopped and have hot fluids in lines, tubes and compartments.

Battery electrolyte contains acid – avoid contact with skin or eyes.

Perform all running checks in an open area free of obstruction and personnel

Ensure the box is locked in the raised position and the steering lock is in place before working on the machine

Ensure the box is free of muck before working on the machine

If working on air system brakes, bleed down air system

Ensure moldboard and ripper are lowered to the ground before servicing equipment

Refer to manufacturer's operation and maintenance manuals for equipment specifications, fluid capacities, operating procedures and maintenance procedures

Be sure the area around the machine is clear of personnel and obstructions while performing running checks

FOR MORE COMPLETE LIST OF SAFETY MESSAGES, REFER TO THE OPERATION AND MAINTENANCE MANUAL FOR THIS MACHINE

X = OK, N = DEFECT FOUND AND REPAIRED, C = DEFECT FOUND **NOT REPAIRED**, R = N/A

		X	N	C	R
⚠	**ALL SAFETY DEVICES AND WARNING LABELS MUST BE OPERATIONAL AND IN GOOD CONDITION BEFORE OPERATING MACHINE!**				
	PRE-OPERATION SAFETY AND ENGINE START-UP				
1	Wash unit (remove bolts) gap before washing engine area)				
2	Set-up book - check unit has book & ensure book is current & remove				
3	Set-up book comments - review				
4	Check condition of guard walls, ensure they all unfold and lock in				
5	Check "OTHER ENERGY" labels, make sure they are in place and legible				
6	Check cab door - ensure door latch mechanism is operational, door hinges are in good condition, ensure door seal is in good condition, check hinge & gas strut				
7	Check condition & operation access steps, lids & prop rods, certify				
8	Wash all boarding surfaces are in good shape and the steps and grab handles are in good shape				
9	Check front & rear frames for cracks & report damage				X
10	Check the frame rub bars for excess wear				
	OPERATING PATTERN				
11	Check proper operation of the dead stick buzzer, certify both and backup alarm				X
12	Test the operation of the teacher lockout (try to start machine while in teach mode)				X
13	Test the operation of the function lockout				X
14	Gauges, dash lights, marker lights, steering lights - check condition & operation				
15	Operator instruct labels - ensure they are in place & legible				
16	Check condition of operator seat (ensure seat is operating accurately and ensure the rubber steps are in good condition)				
17	Labels - inspect labelling on unit (CSM, warning, operator labels)				
18	Brake test procedure please - check if it is in place				
19	Brake pedal - check for smooth operation, wear of pivot & remove any obstructions				X
20	Ensure CB is functioning properly				X
21	Service brake holding capacity - test				X
22	Test brake holding capacity - test				X
23	Throttle - check for smooth operation & remove any obstructions				X
24	Accelerator - check for smooth operation & remove any obstructions on floor				X
25	Surface RCM download, provide download file to Flamanco / Reliability				
26	Seat and armrests - check overall condition and ensure the height and adjustment functions are in place				
27	Seatbelt - check condition & expiry date				X
28	ROPS / FOPS - check for damage & loose mounting bolts				
29	Hydraulic & transmission control levers - check for free movement & looseness				X
30	Check condition of the door, hinges, latches, locking-setup, and door seals				
31	Oil leaks in cab & under door panel - check for leaks				
32	Check condition of accessorie stepes and certify the steps and markers are working				
33	Fire extinguisher - check condition and ensure the emergency egress hab is in place				X
34	Check operation of the heating / air conditioning system including the cab pressurization system, ensure the cab door seals are in good condition (test and the directional vents are pressure and in good condition				
	ENGINE COMPT				
35	Engine breather tube - check for excessive smoke or leaking engine oil				
36	Transmission - check for oil leaks				
37	Exhaust / Fission - operate unit, listen for noise and ensure that all fasteners are tightened				X
38	Perform exhaust consumber safty and call test				X
39	Verify operation of the bumpo lockup				
40	Verify all hydraulic functions are working, including flow				
41	Check operation of the payload monitoring system, calibrate system				
42	Ensure the box on goal lockout is working correctly				
43	Verify operation of the heavy exhaster / compression maps				X
44	Check for play in the articulation pins, ensure that all of the bolts are nuggs				X
45	Hydraulic hose - check clearance hold maps to avoid chafeing cylinders, hydraulic hot, adjust chafeing straps and nuggs(list cylin if				
46	Check steering cycle times at high idle (low)	Left to Right:	Right to Left:		■
47	Verify secondary (Emergency) steering is operational				
48	Check condition of steering cylinder, ensure the steering cylinder pins and bushings, Ensure pin retainers are tight and in place				
49	Check for play (including side to side play) in the left cylinder pins and bushings, ensure pin retainers are tight and in place				
50	Check lift cylinder cycle times (low)	Raised to fully raised:	Fully raised to raised:		■
	Check left cylinder drift test				
	Check the operation of the backup camera, ensure the camera is aimed correctly Reference pylon 3 under camera, pylon 2 at 6' (2m) directly in line with 5, and pylon 1 directly in line with 2 and 3 as far away as possible Position pylons 4 and 5 at 4' (1m)to the right and left of the camera centerline respectively and halfway between pylons 1 and 2 Position pylons 6 and 7 at 2' (0.5m) to the left and right respectively with the camera Note: Position as shown in illustration. Camera view (illustration 2) shows the view from the camera. Pylons 1 and 2 must be in the camera view, Pylons 4, and 5 must be the same distance to the outside of the monitor as per arrows in camera view, Pylons 6 and 7 must not be seen by the camera 				
51	Perform thermal engineerinc				
	ISOLATION				
52	Lock machine and place on jack stands (or rumpo)				
53	Bleed hydraulic and brake pressures				
54	Install steering lock and boom lock				
55	Tag and lockout machine				
56	Spectrum energy area				
57	Ensure cabin tilt function is working properly and that the cylinder is in good condition				
58	Install cabin tilt lock				
	OIL SAMPLES + OIL AND FILTER CHANGES				
59	Take engine oil sample, change oil, inspect magnetic plug				■
60	Note magnetic plug Item 1: 0 Based on "Magnetic Plug Ratings"				■
61	Replace engine oil filter, cut old filter and inspect				■
62	Note filter debris Item 2: 0 based on "Filter Contamination Table"				■
63	Replace water separator & fuel filter				
64	Replace engine fuel filter				
65	Engine coolant - take sample, check coolant level				
66	Replace engine coolant filter				
67	Transmission oil - take sample, check oil level				
68	Change transmission filter, cut old filter and inspect				■
69	Note filter debris Item 3: 0 Based on "Filter Contamination Table"				■
70	Gearbox oil - take sample, check oil level				
71	Change gearbox filter, cut old filter and inspect				■
72	Note filter debris Item 4: 0 Based on "Filter Contamination Table"				■
73	Replace (treatment) fumes disposal filter (AIS)				
74	Clean ARC filter screen				
75	Hydraulic system - take sample, check oil level				
76	Change hydraulic (return) filter, cut old filter and inspect				■
77	Note filter debris Item 5: 0 Based on "Filter Contamination Table"				■
78	Change brake filter, cut old filter and inspect				■
79	Note filter debris Item 6: 0 Based on "Filter Contamination Table"				■
80	Front and rear differential oil - take sample, change oil, check magnetic plugs				■
81	Note magnetic plug Item 1: 0 Based on "Magnetic Plug Ratings"				■
82	Take oil sample, change final drive oil, inspect magnetic plugs				■
83	Note magnetic plug Item 2: 0 Based on "Magnetic Plug Ratings"				■
84	Take sample and change final drive oil (all four corners), inspect magnetic plugs				
85	Complete proper labelling and packaging of samples ready to be sent out for analysis				
86	Replace primary & secondary air filters				
87	Replace cab air filter				
88	Replace recirculating air (A/C) filter				
	ENGINE COMPARTMENT				
89	Inspect - check condition & tightness of mount bolts, inspect fins for damage and certify the fins are not plugged and the tubes are not leaking Inspect - check condition & tightness of mount bolts, inspect fins for damage and certify the fins are not plugged and the tubes are not leaking Inspect - check condition & tightness of mount bolts, inspect fins for damage and certify the fins are not plugged and the tubes are not leaking Inspect - check condition & tightness of mount bolts, inspect fins for damage and certify the fins are not plugged and the tubes are not leaking AIR CONDITIONING CONDENSER - check condition & tightness of mount bolts, inspect fins for damage and certify the fins are not plugged and the tubes are not leaking, and ensure fans are secured to the condensor body and welding				
90	Inspect - check condition of overall system, including filter housing and pre-cleaner, piping to and from engine and aftercooler, and the mounting bracket, check for any signs of damage, check for any air and rubber hoses for degradation, and all components				X
91	Inspect - check condition of overall system including piping, flange, purifiers and exhaust snap on piping				
92	Check condition of exhaust system including clamps (loose clamps may be a quick indicator of internal leaks) and check for any evidence of oil coming from the intake and exhaust housings				X
93	Batteries - check all terminals and posts are free from corrosion and are tight				
94	Check level of electrolyte in the battery and test specific gravity (if battery is not maintenance free)				
95	Check for loose or corroded wiring on battery and master disconnect switch				
96	Check condition of all wiring look condition & routing				
97	Check starter mounting bolts and wiring connections				
98	Check for loose or corroded wiring on starter				
99	Check alternator mounting bolts and wiring				
100	Check for loose or corroded wiring for alternator				
101	Check belt tension and condition				X
102	Check condition of all hoses for leaks				
103	Inspect fuel lines for leaks and rissaurs, inspect fitment of all clamps				
104	Check for leaks from metal pump, hoses, radiator and overall cleanliness of cooling system				
105	Check condition of engine covers and check all guards are in place & secure				
106	Oil & fuel leaks - check engine bay for leaks & cleanliness				
107	Hydraulic, transmission and hoses/cooling pump - check for leaks				
	TIRES				
108	Check tires for damp cutters bouncy from the tyre, face cutting, lump cuts (cords exposed), sidewall damp (depart damage), sidewall rain (cords exposed), or any tyre bulging Check tires for damp cutters bouncy from the tyre, cracking or loose studs / nuts, elongated bolt holes, damage to rims look ring Please record serial numbers from the tires LF: [ ] RF: [ ] LR: [ ] RR: [ ] LO: [ ] RO: [ ] LI: [ ] RI: [ ] Measure inside & outside tire depth, record lowest reading (in/mm) LF: [ ] RF: [ ] LR: [ ] RR: [ ] LO: [ ] RO: [ ] LI: [ ] RI: [ ] Measure diameter with tape measure across 2 tires (lowest where to is more than 2% difference side to side and 4% difference front to back) LF: [ ] RF: [ ] LR: [ ] RR: [ ] LO: [ ] RO: [ ] LI: [ ] RI: [ ] Check inflation pressure in tires LF: [ ] RF: [ ] LR: [ ] RR: [ ] LO: [ ] RO: [ ] LI: [ ] RI: [ ]				X X ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■
	PODESTRIAN				
109	Check condition of transmission mounts & torque mounting bolts				X
110	Inspect all universal joints for wear and play, check & tighten bolts				X
111	Check tightness of ventilating axle bolts, check for play in the ventilation bearings				
112	Check front, center, and rear axle mounting frame pivot pins and bushings, check for play in the pivot points				X
113	Check front center, and rear, stabilizer lines for excess play or wear				X
114	Check suspension cylinders for leaks or damage, measure suspension height to ensure the cylinder is still in good condition				X
115	Check condition of all mounting bolts using hammer (tapping) to identify any loose bolts				
116	Check torque / installation compatibnch for any leaks, check hoses for wear, damage and proper routing, check electrical harnesses for proper routing and for any damage				
117	Check condition of ground strap in the articulation joint				
	DUMP BODY / DIRECTOR BODY				
118	Check box for wear and cracks				
119	Check for excess wear or gaps on box steel support pads				
120	Ensure grease hoses are in good condition and all pins are having				X
121	Inspect condition dump cylinders				
	HYDRAULIC AND HYDRAULIC TANK COMPARTMENT				
122	Check condition & routing of all hoses in hydraulic tank bay				
123	Check condition & routing of all hoses in articulation area				
124	Check condition & routing of all hoses in rear frame area				
125	Check condition & installation of hydraulic sensor				
126	Check hydraulic hose condition & routing, check for leaks				
127	Check all valves for leaks				
	GREASING				
128	Check level of the autogreaser reservoir				
129	Check for proper operation of grease pump				X
130	Check for leaks at all lines & connections on auto lube system, if applicable				X
131	Ensure for bearing on engine				
132	Check for proper grease distribution at all auto lube points				
133	Grease all universal joints, drip joints & pillow block				
134	Grease all cover and dump hinges				
	POST WORK TASKS				
135	Perform all final tasks and verify all dones and boxes ships are				
136	Take care ping closure, service panel, lower, and door to properly, thread and lubed Clean work area of debris, old parts, excess / unused materials and lubricants, mill up boxes and cards, return chemicals to flammable storage				
137	Return equipment parked up or inspected to a safe operating condition				
138	Ensure work orders are submitted to planning for all found corrective issues to be resolved at a later date				
139	Ensure all tools and equipment is signed off, dated and returned to Supervisor / Foreman				
140	Refit around equipment to ensure that all foreign materials are cleaned up & put away properly. (bags, tools, sticks, chalks, parts, garbage)				
141	Ensure all consumables are accounted for and signed off on the job card, Hand made soap sheets are filled out & handed into supervisor for review				
142	Complete parts & core returns				
143	Clean up old filters / oils and dispose of properly				
	RE-INSTALLATION				
144	Remove lock pin and lower cable, ensure it is secured				
145	Remove pedlock & danger tags & re-energize equipment				
146	Remove articulation lock, tie up lock, lower equipment to the ground and remove footlocks				
	START-UP				
147	Check work and inspect for any fluid leaks				
148	Verify all fluids levels				
149	Verify operations done made is completed				

SIGN OFF

TECHNICIAN NAME:

TECHNICIAN SIGNATURE:

SUPERVISOR NAME:

SUPERVISOR SIGNATURE:

DATE:

DEFFICIENCY LIST
CATERPILLAR SCOOP (LHD)
HR - PM

Step No.	Defficiency	Corrective Action Required	Work Request No.	Comp

PM FEEDBACK FORM

HAVE YOU FOUND ANY PARTS DISCREPANCIES?			
STOCK #	***INCORRECT PART #***	***CORRECT PART #***	***DESCRIPTION OF PART***

HAVE YOU IDENTIFIED ANY MAINTENANCE TASK TO BE UPDATED, ADDITION OR DELETED?	
TASK DESCRIPTION:	***UPDATE / ADD / DELETE (circle one)***